

GRANT COUNTY REGIONAL, OR, USA

WMO: 726876

Lat: 44.400N

Lon: 118.967W

Elev: 1127

StdP: 88.50

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00387

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.2	-8.9	-16.0	1.1	-8.7	-12.9	1.4	-4.3	14.3	7.2	12.3	6.2	2.6	270	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	34.9	15.9	33.0	15.5	31.6	15.1	17.7	30.7	16.7	29.7	15.9	28.7	3.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.3	10.9	19.1	12.3	10.2	18.7	11.2	9.5	18.5	54.2	30.7	51.0	29.4	48.6	29.0	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	8.5	7.3	-16.2	37.7	2.9	1.9	-18.2	39.1	-19.9	40.2	-21.5	41.3	-23.6	42.6		
(5)			-16.8	19.2	3.4	1.2	-19.2	20.1	-21.2	20.8	-23.0	21.4	-25.5	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	1.7	2.9	5.4	8.0	12.4	16.6	22.2	21.5	17.1	10.5	5.0	0.4	(6)
(7)		DBStd	8.44	4.45	4.29	3.76	3.93	4.27	4.26	3.08	3.36	4.45	4.26	4.90	4.84	(7)
(8)		HDD10.0	1223	258	199	147	84	24	3	0	0	3	46	161	297	(8)
(9)		HDD18.3	3218	516	431	399	309	188	81	6	12	74	246	401	555	(9)
(10)		CDD10.0	1358	1	1	6	26	99	202	378	357	217	61	11	1	(10)
(11)		CDD18.3	312	0	0	0	0	5	29	127	111	38	2	0	0	(11)
(12)		CDH23.3	4658	0	0	1	18	125	498	1806	1564	606	40	0	0	(12)
(13)	CDH26.7	2103	0	0	0	1	28	181	894	748	240	11	0	0	(13)	
(14)	Wind	WSAvg	2.9	3.2	3.2	3.3	3.1	2.7	2.6	2.6	2.6	2.8	3.2	3.2	(14)	
(15)	Precipitation	PrecAvg	375	41	27	34	41	43	35	14	14	31	42	42	(15)	
(16)		PrecMax	500	77	59	78	67	99	70	43	52	42	60	75	104	(16)
(17)		PrecMin	228	13	11	16	22	1	4	1	0	1	11	13	15	(17)
(18)		PrecStd	71	19	11	17	13	26	18	13	13	9	12	18	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.7	19.8	26.1	29.8	34.7	36.8	37.2	34.1	28.0	20.1	12.9	(19)
(20)			MCWB	6.3	7.4	9.5	12.3	16.0	17.7	16.7	16.6	15.3	13.1	10.7	7.6	(20)
(21)		2%	DB	11.0	13.3	16.6	22.4	27.0	31.2	35.1	34.8	31.6	23.0	16.8	11.0	(21)
(22)			MCWB	5.6	6.9	7.8	11.2	14.0	16.2	15.9	15.7	14.5	11.3	9.2	6.4	(22)
(23)		5%	DB	9.0	11.2	14.2	19.0	24.0	28.3	33.7	32.7	29.8	20.7	14.3	8.7	(23)
(24)			MCWB	5.1	6.1	7.0	9.7	12.9	14.9	15.7	15.2	14.0	10.4	8.2	5.4	(24)
(25)		10%	DB	7.6	9.1	12.3	15.9	21.5	26.2	32.1	31.1	27.1	17.8	12.1	7.1	(25)
(26)			MCWB	4.2	5.0	6.6	8.2	11.8	13.8	15.5	14.8	13.0	9.6	7.2	4.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	8.8	10.6	13.3	16.7	19.0	19.1	18.1	16.9	14.2	11.6	8.3	(27)
(28)			MCDWB	11.5	13.6	17.1	23.7	28.1	31.5	33.0	32.4	28.3	23.5	17.9	11.9	(28)
(29)		2%	WB	6.5	7.7	9.1	11.6	14.9	17.2	17.9	16.9	15.8	12.7	10.3	6.9	(29)
(30)			MCDWB	9.5	12.4	14.4	21.0	23.8	28.7	31.5	30.5	28.3	20.0	15.3	10.1	(30)
(31)		5%	WB	5.4	6.5	8.0	10.2	13.7	15.8	16.8	16.1	14.8	11.4	9.0	5.7	(31)
(32)			MCDWB	8.4	10.3	13.2	17.7	21.9	26.5	30.4	29.6	26.8	18.3	13.7	8.3	(32)
(33)		10%	WB	4.4	5.3	6.9	8.9	12.6	14.6	16.0	15.4	13.7	10.5	7.6	4.3	(33)
(34)			MCDWB	7.2	8.9	11.7	14.7	19.8	24.3	29.4	28.7	25.3	16.4	11.5	6.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	10.1	11.5	13.1	14.4	17.6	17.4	15.7	11.7	8.6	6.8	(35)
(36)			MCDBR	9.6	11.1	13.7	17.4	17.3	18.3	19.4	19.6	19.1	16.4	12.0	7.8	(36)
(37)			MCWBR	5.9	6.3	7.2	8.4	7.6	7.2	7.0	7.0	7.5	7.5	6.4	4.9	(37)
(38)		5% WB	MCDBR	8.3	9.9	11.7	15.7	15.2	16.9	17.7	17.8	17.5	13.6	10.6	7.1	(38)
(39)			MCWBR	5.6	6.0	6.6	8.0	6.9	6.8	6.9	6.5	7.0	6.9	6.0	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.282	0.286	0.298	0.350	0.355	0.340	0.346	0.344	0.325	0.301	0.283	0.279	(40)
(41)		taud		2.487	2.531	2.527	2.353	2.375	2.449	2.451	2.449	2.487	2.563	2.557	2.511	(41)
(42)		Ebn at Noon		852	918	947	912	914	926	917	908	902	886	849	821	(42)
(43)		Edh at Noon		69	79	90	117	118	110	109	105	93	75	64	61	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.57	2.47	3.63	5.13	6.10	7.17	7.71	6.60	5.00	3.17	1.79	1.23	(44)
(45)		RadStd		0.20	0.28	0.38	0.43	0.52	0.40	0.30	0.38	0.32	0.27	0.16	0.18	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page